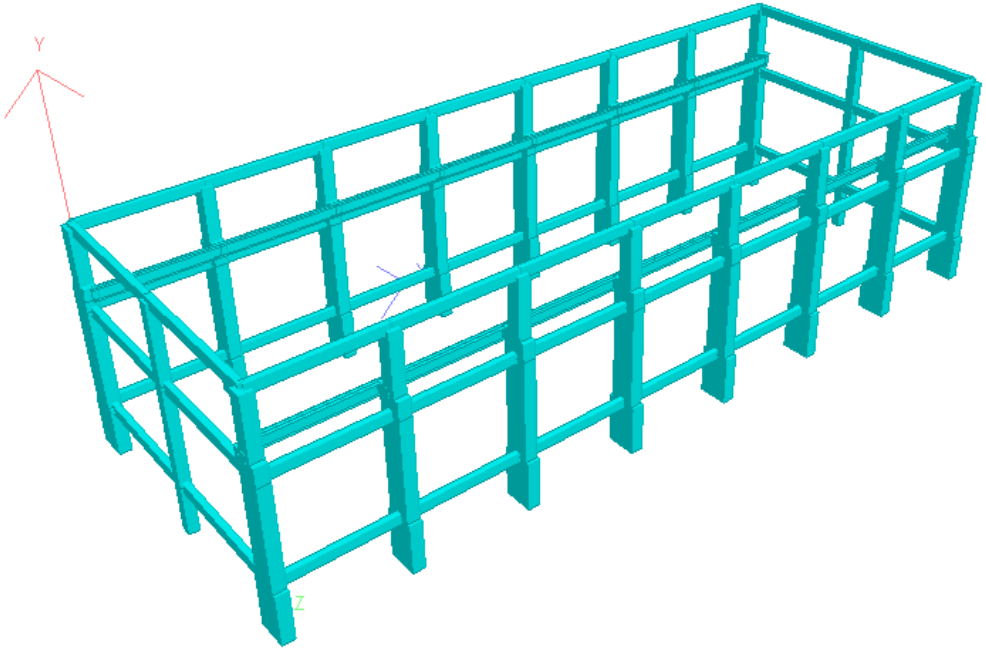
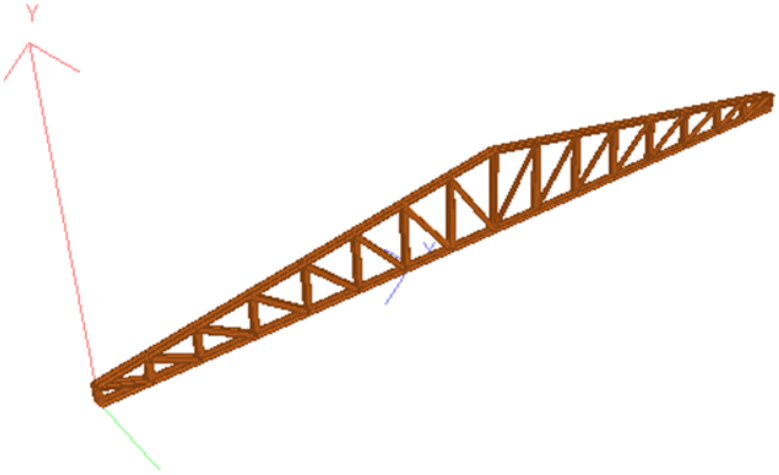

Y
Z


Y

GENERAL NOTES

1. THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND CONDITIONS AT SITE AND SHALL NOTIFY THE ENGINEER ON RECORD OF ANY DISCREPANCY BETWEEN THE ACTUAL CONDITION AND THE INFORMATION SHOWN ON THE DRAWINGS. THESE SHALL INCLUDE BUT NOT LIMITED TO THE LOCATIONS AND DIMENSIONS OF GROOVES, CHASES, SLEEVES, CURBS, OPENINGS, EMBEDDED OR ATTACHED ITEMS, ETC. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING DRAWINGS.
2. IN THE INTERPRETATION OF THESE DRAWINGS, INDICATED DIMENSIONS SHALL TAKE PRECEDENCE OVER SCALED MEASUREMENTS OF PLANS, SECTIONS AND DETAILS. NOTES AND DETAILS ON DRAWINGS SHALL TAKE PRECEDENCE OVER THE GENERAL NOTES AND THE TYPICAL DETAILS.
3. THE STRUCTURAL DRAWINGS HEREIN REPRESENT THE FINISHED STRUCTURE. THEY DO NOT INDICATE THE METHOD OF CONSTRUCTION UNLESS SO STATED. THE CONTRACTOR SHALL THEREFORE PROVIDE ALL THE NECESSARY MEASURES IN ORDER TO PROTECT THE STRUCTURE, THE ADJACENT PROPERTIES, WORKMEN AND ALL OTHER PERSONS AT SITE DURING ALL PHASES OF THE CONSTRUCTION. THE CONTRACTOR SHALL IMMEDIATELY NOTIFY THE ENGINEER ON RECORD OF ANY CONDITION WHICH IN HIS OPINION MIGHT ENDANGER THE STABILITY OF THE STRUCTURE OR MIGHT CAUSE DISTRESS TO THE STRUCTURE.
4. CONSTRUCTION MATERIALS SHALL NOT BE STORED ON POURED SLABS. IT IS THE RESPONSIBILITY OF THE GENERAL CONTRACTOR TO ENSURE THAT THE SUB-CONTRACTOR/S (IF THERE ARE ANY) ARE INFORMED OF THIS SO AS NOT TO VIOLATE THIS REQUIREMENT.
5. THE CONTRACTOR SHALL PROVIDE ALL TEMPORARY GUYING AND/OR BRACING REQUIRED TO ERECT AND HOLD THE STRUCTURE IN PROPER ALIGNMENT UNTIL ALL STRUCTURAL WORKS AND CONNECTIONS HAVE BEEN COMPLETED. THE INVESTIGATION, DESIGN, SAFETY, ADEQUACY AND INSPECTION OF ERECTION BRACING, SHORING, TEMPORARY SUPPORTS, ETC. ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR.
6. THE CONTRACTOR SHALL TAKE ALL THE NECESSARY STEPS TO ENSURE THE PROPER ALIGNMENT OF THE STRUCTURE AFTER THE INSTALLATION OF ALL STRUCTURAL AND FINISHED MATERIALS.
7. UNLESS OTHERWISE SHOWN, ALL DIMENSIONS ARE IN METERS.
8. ALL COORDINATES AND ELEVATIONS ARE IN METERS.
9. ALL CONCRETE WORK SHALL BE DONE IN ACCORDANCE WITH THE LATEST ACI BUILDING CODE FOR REINFORCED CONCRETE AND ALL STRUCTURAL STEEL WORK SHALL BE IN ACCORDANCE WITH THE LATEST AISC SPECIFICATIONS IN SO FAR AS THEY DO NOT CONFLICT WITH THE LOCAL BUILDING REQUIREMENTS.

CONCRETE

1. UNLESS OTHERWISE INDICATED IN PLANS OR NOTED IN THE SPECIFICATIONS THE MINIMUM 28-DAY CYLINDER COMPRESSIVE STRENGTH OF CONCRETE SHALL BE AS FOLLOWS:

1. FOOTINGS, FOOTING TIE BEAMS AND SLAB ON GRADE	f'c = 21.00 MPa (3000 psi)
2. PEDESTAL, COLUMNS AND R.C. WALLS	f'c = 21.00 MPa (3000 psi)
3. R.C. FLOOR SLABS, BEAMS AND GIRDERS	f'c = 21.00 MPa (3000 psi)
4. PARAPET AND OTHER STRUCTURAL ELEMENTS	f'c = 21.00 MPa (3000 psi)
5. LEAN CONCRETE	f'c = 14.00 MPa (2000 psi)

2. AGGREGATE SIZE SHALL BE AS FOLLOWS:

1. COLUMNS, GIRDERS, BEAMS, SLABS AND FOOTINGS	20 mm MAXIMUM
2. FOOTINGS AND SLAB ON GRADE	40 mm MAXIMUM

3. MINIMUM CONCRETE COVER FOR REINFORCING BARS SHALL NOT BE LESS THAN THE FOLLOWING:

1. UNFORMED CONCRETE DEPOSITED AGAINST THE GROUND	75 mm
2. FORMED CONCRETE EXPOSED TO GROUND OR WEATHER WHEN BARS ARE GREATER THAN ø16 MM	50 mm
3. FORMED CONCRETE EXPOSED TO GROUND OR WEATHER WHEN BARS ARE ø16 MM OR SMALLER	40 mm
4. BEAMS OR COLUMNS NOT EXPOSED TO GROUND OR WEATHER	40 mm
5. SUSPENDED SLABS AND WALLS NOT EXPOSED TO GROUND OR WEATHER	20 mm
6. CONCRETE EXPOSED TO SEWAGE	75 mm

4. ALL REINFORCING STEEL, DOWELS, ANCHOR BOLTS AND OTHER INSERTS SHALL BE SECURED IN POSITION PRIOR TO POURING OF CONCRETE.

REINFORCING BARS

1. ALL REINFORCING BARS SHALL BE FABRICATED AND INSTALLED IN CONFORMANCE WITH THE MANUAL OF STANDARD PRACTICE FOR DETAILING REINFORCED CONCRETE STRUCTURES, ACI 315 LATEST EDITION.
2. ALL REINFORCING STEEL BARS SHALL BE NEW BILLET, HOT ROLLED, DEFORMED BARS CONFORMING TO THE SPECIFICATIONS OF PNS 49:2002 WHOSE MINIMUM YIELD STRENGTH SHALL BE 275 MPA (GRADE 40) FOR RSB 12mmø AND 414 MPA (GRADE 60) FOR RSB 16mmø OR GREATER.
3. ALL REINFORCING BARS SHALL BE ACCURATELY AND SECURELY PLACED BEFORE POURING CONCRETE OR GROUT OR BEFORE APPLYING MORTAR.
4. ALL REINFORCING BARS SHALL BE CLEAN OF RUST, GREASE OR OTHER MATERIAL THAT MAY IMPAIR BOND.

STRUCTURAL STEEL

1. ALL STRUCTURAL STEEL SHALL HAVE A MINIMUM YIELD STRENGTH OF Fy = 248 MPa AND CONFORM WITH ASTM A36 SPECIFICATION.
2. ALL STRUCTURAL STEEL SHALL BE FABRICATED AND ERECTED IN ACCORDANCE WITH THE AISC SPECIFICATIONS AND CODE OF STANDARD PRACTICE.
3. ALL ANCHOR BOLTS, NUTS AND WASHERS, TURNBUCKLES AND TENSION BARS SHALL HAVE A STRENGTH OF ASTM A307.
4. SHOP AND FIELD WELDINGS SHALL BE IN ACCORDANCE WITH AWS D1.1 AND SHALL BE PERFORMED BY QUALIFIED WELDERS.
5. UNLESS OTHERWISE INDICATED, WELDING ELECTRODES SHALL BE E70XX. THE MINIMUM THICKNESS OF WELD SHALL BE 3MM.
6. THE CONTRACTOR SHALL FURNISH ALL PLATES, CLIP ANGLES, SPACERS, CONNECTORS, ETC. REQUIRED FOR THE COMPLETION OF THE WORK EVEN IF SUCH ARE NOT SHOWN IN THE DRAWINGS.

MASONRY

1. ALL MATTERS AND WORKMANSHIP SHALL BE IN ACCORDANCE WITH THE APPLICABLE STANDARDS AND SPECIFICATIONS OF THE NATIONAL CONCRETE MASONRY ASSOCIATION AND UNIFORM BUILDING CODE.
2. CONCRETE MASONRY UNITS SHALL CONFORM TO ASTM C90 GRADE N.
3. UNLESS OTHERWISE INDICATED, CHB IN THIS WORK SHALL HAVE A MINIMUM ULTIMATE COMPRESSIVE STRENGTH AT 28 DAYS OF Fm = 2.7 MPa (400 psi).
4. CONCRETE MASONRY UNITS SHALL BE FILLED SOLIDLY WITH MORTAR. CONCRETE MORTAR AND GROUT FOR ALL REINFORCED MASONRY SHALL CONFORM TO ASTM 270 TYPE M AND SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 10.00 MPa (1500 psi) AT 28 DAYS.
5. PROVIDE A SOLID BAND OF CONCRETE LINTEL BEAM AT THE TOP OF A CHB WALL THAT HAS NO UPPER SUPPORT. THE LINTEL BEAM SHALL HAVE A WIDTH EQUAL TO THE THICKNESS OF THE WALL AND A DEPTH OF NOT LESS THAN 200 MM.
6. ALL MASONRY WALLS SHALL BE REINFORCED ACCORDING TO THE FOLLOWING SCHEDULE OF CONCRETE HOLLOW BLOCK REINFORCEMENT UNLESS OTHERWISE INDICATED IN THE PLANS.

SLAB ON GRADE

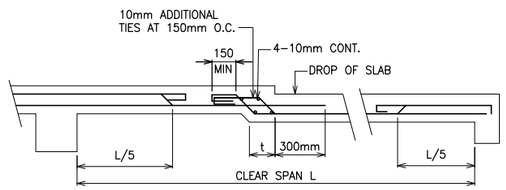
1. THE SOIL SUBGRADE AND FILL LAYERS BELOW ALL SLABS ON GRADE, PAVING AND PIT SLABS SHALL BE MECHANICALLY COMPACTED IN LAYERS TO A MINIMUM OF 95 PERCENT OF THE MODIFIED PROCTOR DENSITY, PER ASTM D1557.
2. ALL SLABS ON GRADE SHALL BE PROVIDED WITH A MINIMUM OF 100 MM THICK COMPACTED CLEAN COARSE SAND BED.
3. UNLESS OTHERWISE NOTED, ALL BEDDED SLABS SHALL BE REINFORCED WITH 12 MM BARS AT 250 MM O.C. EACHWAY AT THE CENTER OF THE SLAB.

COLUMN

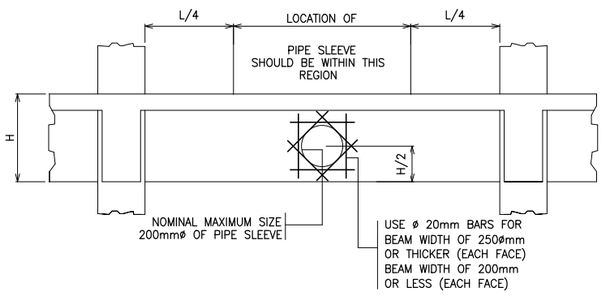
1. WHERE COLUMNS CHANGE IN SIZE, VERTICAL REINFORCEMENTS SHALL BE OFFSET AT A SLOPE NOT MORE THAN 1 ON 6. PROVIDE TRANSVERSE REINFORCEMENT FOR JOINTS WITH BAR OFFSET.
2. LAP SPLICES WHEN REQUIRED, ARE PERMITTED ONLY WITHIN THE CENTER OF THE COLUMN LENGTH AND SHALL BE PROPORTIONED AS TENSION SPLICES. IN NO CASE SHALL THE LAP SPLICE BE LOCATED CLOSER THAN A DISTANCE EQUAL TO THE MAXIMUM COLUMN DIMENSION FROM THE FACE OF THE BEAM-COLUMN JOINT. PROVIDE EXTRA TRANSVERSE REINFORCEMENT OF THE SAME SIZE AND ARRANGEMENT INDICATED IN THE COLUMN SCHEDULE SPACED AT MOST ONE-FOURTH THE MINIMUM COLUMN SECTION DIMENSION THROUGHOUT THE LENGTH OF THE SPLICE OR 100 MM.
3. FOR ALL TIED COLUMNS, PROVIDE TRANSVERSE REINFORCEMENT OF THE SAME SIZE AND ARRANGEMENT INDICATED IN THE COLUMN SECTION SCHEDULE AND SPACED NO GREATER THAN ONE-QUARTER THE MINIMUM COLUMN SECTION DIMENSION NOR 100 MM, OVER A DISTANCE FROM EACH JOINT FACE OF NOT LESS THAN THE LARGER OF THE MAXIMUM COLUMN SECTION DIMENSION, OR ONE-SIXTH OF THE CLEAR HEIGHT OF THE COLUMN OR 450 MM.

BEAM AND GIRDER

1. UNLESS OTHERWISE NOTED IN PLANS OR SPECIFICATIONS, CAMBER ALL BEAMS AT LEAST 8 MM FOR EVERY 450 MM OF SPAN EXCEPT FOR CANTILEVERS FOR WHICH THE CAMBER SHALL BE AS NOTED IN THE PLANS OR AS ORDERED BY THE STRUCTURAL ENGINEER BUT IN NO CASE LESS THAN 19 MM FOR EVERY 3000 MM OF FREE SPAN.
2. IF THERE ARE TWO OR MORE LAYERS OF LONGITUDINAL REINFORCING BARS IN A BEAM OR GIRDER, USE SEPARATORS OF A SIZE NOT LESS THAN 25 MM BARS SPACED ABOUT 900 MM ON CENTER. IN NO CASE SHALL THERE BE LESS THAN TWO (2) SEPARATORS BETWEEN LAYERS OF BARS.
3. LONGITUDINAL REINFORCING BARS SHALL BE PLACED SYMMETRICALLY ABOUT THE VERTICAL CENTER LINE OF THE BEAM OR GIRDER SECTION WHERE POSSIBLE WITH UPPER LAYER BARS PLACED DIRECTLY ABOVE THOSE IN THE BOTTOM LAYER.
4. MINIMUM CONCRETE PROTECTION OF REINFORCING BARS OR SHAPES SHALL BE AS SHOWN IN FIGURE 4. BEAM REINFORCING BARS BOTH TOP AND TERMINATING IN A WALL SHALL EXTEND AT THE MOST 50 MM FROM THE FACE OF THE WALL AND SHALL TERMINATE IN A STANDARD 90°.
5. WHEN A BEAM CROSSES A GIRDER REST BEAM BARS ON TOP OF THE GIRDER BARS. PROVIDE ADDITIONAL TIES WHEN BEAMS OF DIFFERENT TOP ELEVATIONS CROSS A GIRDER.
6. LONGITUDINAL REINFORCEMENTS OF GIRDERS, BOTH TOP AND BOTTOM TERMINATED IN A COLUMN SHALL BE EXTENDED TO THE FAR FACE OF THE CONFINED CONCRETE CORE OF THE COLUMN AND TERMINATED BY A STANDARD 90° HOOK.
7. GENERALLY, NO LAP SPLICE SHALL BE PERMITTED ON BEAMS AND GIRDERS AT POINTS WHERE CRITICAL BENDING STRESSES OCCUR. IN ADDITION, FOR GIRDERS, NO LAP SPLICES SHALL BE LOCATED WITHIN THE JOINTS OR WITHIN A DISTANCE EQUAL TO TWICE THE MEMBER DEPTH FROM THE FACE OF THE JOINT.
8. PROVIDE LAP SPLICES IN GIRDERS WITH HOOP REINFORCEMENT OVER THE LENGTH OF THE LAPPED BARS SPACED NO FARTHER THAN ONE-FOURTH THE NOMINAL DEPTH, OR 100 MM.
9. WELDED SPLICES AND MECHANICAL CONNECTIONS MAY BE USED FOR SPLICING BEAMS AND GIRDERS PROVIDED THAT NO MORE THAN ALTERNATE BARS IN EACH LAYER OF LONGITUDINAL REINFORCEMENT AND SPLICES AT A SECTION AND THE CENTER TO CENTER DISTANCE BETWEEN SPLICES OF ADJACENT BARS IS AT LEAST 600 MM MEASURED ALONG THE AXIS OF THE MEMBER.
10. SEE MECHANICAL, PLUMBING, ELECTRICAL AND FIRE PROTECTION DRAWINGS FOR ALL SUSPENDED AND EMBEDDED PIPING, CONDUITS, DUCTWORKS, EQUIPMENTS, ETC.
11. PIPE AND DUCT SLEEVES SHALL BE LOCATED WITHIN THE REGION BOUNDED BY ONE-FOURTH OF CLEAR SPAN LENGTH FROM THE SUPPORTS.



1 DROP SLAB
NTS



2 TYPICAL PIPE SLEEVE DETAIL
NTS

FOUNDATION

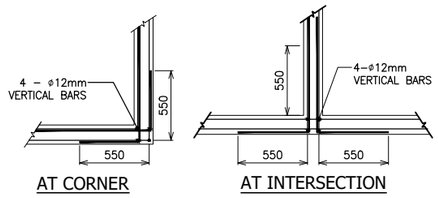
1. FOOTINGS ARE DESIGNED FOR AN ALLOWABLE SOIL BEARING CAPACITY OF 434 KPa AT A MINIMUM OF 3.00 METERS FROM THE NATURAL GRADE LEVEL. THE CONTRACTOR SHALL REPORT IN WRITING TO THE STRUCTURAL ENGINEER ON THE ACTUAL SOIL CONDITIONS UNCOVERED AND CONFIRM ACTUAL BEARING CAPACITY OF SOIL BEFORE DEPOSITING CONCRETE.
2. PROVIDE TEMPORARY REMOVAL OF WATER FROM ANY SOURCE DURING CONSTRUCTION. DEWATERING SHALL BE CAREFULLY AND PROPERLY PERFORMED TO AVOID DISTURBING THE FOUNDATIONS AND SLAB BEARING SURFACES.
3. THE CONTRACTOR SHALL DESIGN, INSTALL AND MONITOR EXCAVATIONS RETENTION SYSTEMS AS REQUIRED FOR PROTECTION OF ADJACENT PROPERTIES/STRUCTURES AND PROVIDE ALL MEASURES AND PRECAUTIONS NECESSARY TO MINIMIZE SETTLEMENT AND PREVALENT DAMAGE TO ADJACENT EXISTING OR NEW CONSTRUCTION.
4. PREPARE CONDITIONS OF CONCRETE SUPPLY AND PLACEMENT OF THE COMPLETE FOUNDATION FOR FULL THICKNESS AS A CONTINUOUS MONOLITHIC CASTING.
5. REFER TO ARCHITECTURAL, MECHANICAL, PLUMBING, ELECTRICAL AND OTHER TRADES FOR SUBSOIL DRAINAGE SYSTEM, MACHINERY ANCHORS AND OTHER EMBEDDED ITEMS, DEPRESSIONS, FINISHES, DOWELS FOR MASONRY WALLS, CURBS, ETC.

CONCRETE SLAB

1. ALL SLAB REINFORCEMENTS SHALL HAVE A MINIMUM CLEAR DISTANCE OF 20 MM FROM THE BOTTOM AND FROM THE TOP OF SLABS.
2. UNLESS OTHERWISE DETAILED, FOR CONTINUOUS SLABS WITH THE MAIN REINFORCEMENT RUNNING IN ONE DIRECTION, REINFORCING BARS SHALL BE BENT UP, EXTENDED OR CUT.
3. IF SLABS ARE REINFORCED BOTHWAYS, BARS ALONG THE SHORTER SPAN BE PLACED BELOW THOSE ALONG THE LONGER SPAN AT THE CENTER OF THE SLAB AND BE PLACED OVER THE LONGER SPAN BARS ON AREAS NEAR THE SUPPORTS. THE SPACING OF THE BARS AT THE COLUMN STRIPS SHALL BE APPROXIMATELY ONE AND ONE-HALF (1-1/2) TIMES THAT IN THE MIDDLE STRIPS BUT IN NO CASE GREATER THAN TWO AND ONE-HALF (2-1/2) TIMES THE SLAB THICKNESS OR 450 MM.
4. TEMPERATURE BARS FOR SLABS SHALL BE GENERALLY PLACED NEAR THE FACE IN TENSION AND SHALL NOT BE LESS THAN 0.0025 BT.
5. UNLESS OTHERWISE NOTED, DROP SLABS SHALL BE PROVIDED WITH ADDITIONAL REINFORCEMENT AT THE LOCATION OF DROP.
6. PROVIDE EXTRA REINFORCEMENT FOR CORNER SLAB (TWO ADJACENT DISCONTINUOUS EDGES).
7. SEE MECHANICAL, PLUMBING, ELECTRICAL AND FIRE PROTECTION DRAWINGS FOR ALL SUSPENDED AND EMBEDDED PIPING, CONDUITS, DUCTWORK, EQUIPMENT, ETC.
8. UNLESS OTHERWISE NOTED, EMBEDDED CONDUITS SHALL BE RUN GENERALLY AT MID-BAY AND PARALLEL CONDUITS SHALL BE AT THREE DIAMETERS ON CENTER. CONDUIT SIZE SHALL NOT EXCEED ¼ OF THE SLAB THICKNESS AND SHALL BE LOCATED AT MID THICKNESS OF THE SLAB.

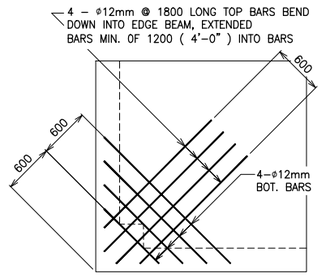
HOLLOW BLOCK REINFORCEMENT SCHEDULE

BLOCK THICKNESS (mm)	REINFORCEMENT		NOTES
	HORIZONTAL	VERTICAL	
100	ø10mm @ 600mm O.C.	ø10mm @ 400mm O.C.	1. MINIMUM LAP SPLICE = 30Db 2. PROVIDE 1-180 VERTICAL BAR AT CORNER INTERSECTION END OF WALLS AND EACH SIDE OF OPENING. 3. DOWELS WITH THE SAME SIZE AS VERTICAL OR HORIZONTAL REINFORCEMENT SHALL BE PROVIDED WHERE CHB WALLS ADJOIN COLUMNS, RC BEAMS AND WALL. 4. LINTEL SHALL BEAR AT LEAST 400mm ON EACH SIDE OF MASONRY WALL OPENINGS.
150	ø12mm @ 600mm O.C.	ø12mm @ 400mm O.C.	
200	ø12mm @ 600mm O.C.	ø12mm @ 400mm O.C.	



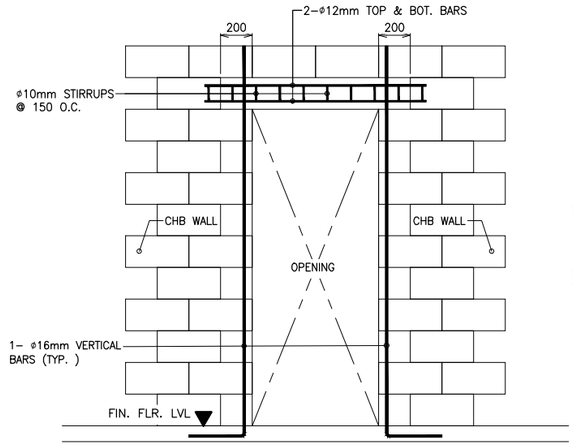
TYP. R.C. WALL CONNECTION DETAIL

SCALE: NTS



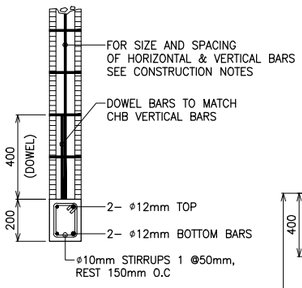
TYP. CORNER SLAB DETAIL

SCALE: NTS

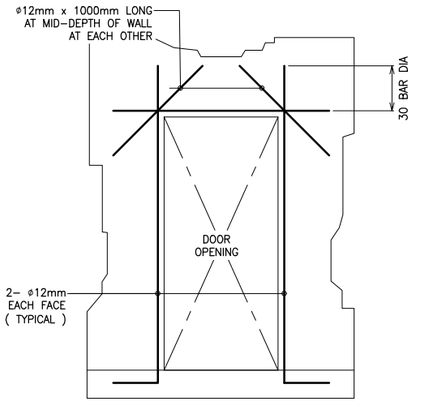


TYP. DETAIL OF LINTEL BEAM AT CHB WALL OPENING

SCALE: NTS

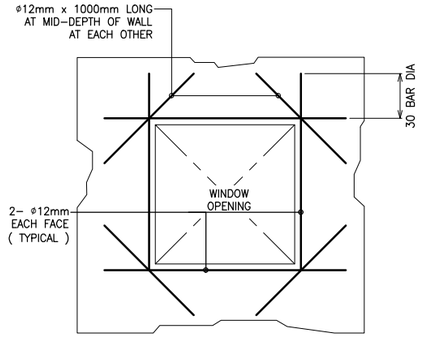


SECTION



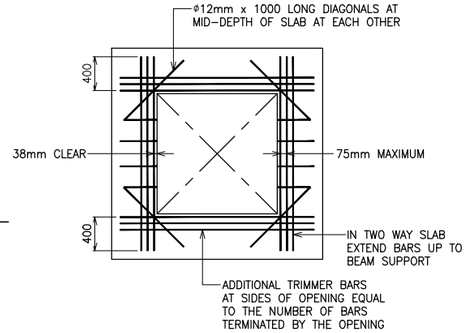
DOOR OPENING DETAIL

SCALE: NTS



WINDOW OPENING DETAIL

SCALE: NTS



NOTE:

OMIT TRIMMER BARS WHERE OPENING IS FRAMED BY BEAMS.

REINFORCING BAR FOR SLAB OPENING	
OPENING SIZE	VERT. & HOR.
UNDER 600	2 – ø12mm
UNDER 800	4 – ø12mm

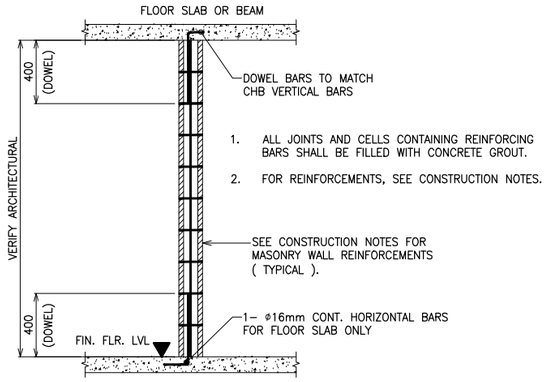
NOTE:

1. PROVIDE THESE ADDITIONAL BARS FOR ALL OPENINGS EXTRA BARS PARALLEL TO THE SIDE OF OPENING EQUAL TO THE NUMBER OF INTERRUPTED BARS BY THE OPENING.

2. SEE ARCHITECTURAL & MECHANICAL PLANS FOR SLAB OPENING LOCATIONS.

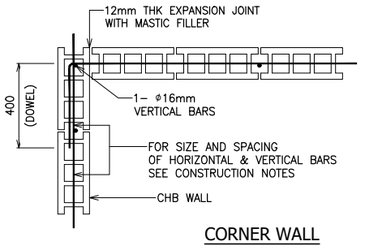
SLAB OPENING DETAIL

SCALE: NTS

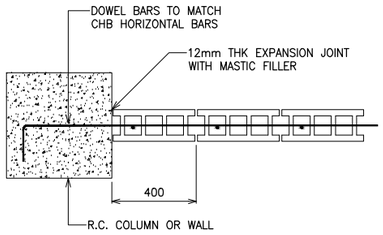


TYP. SECTION OF MASONRY PARTITION REINFORCEMENT

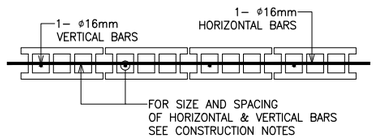
SCALE: NTS

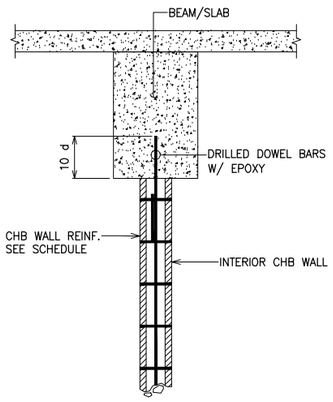


CORNER WALL

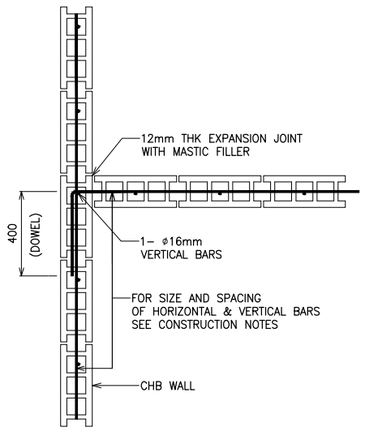


INTERSECTING R.C. COLUMN OR WALL





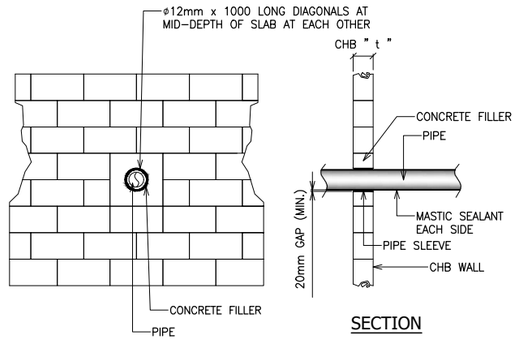
OPENING OR END WALL



INTERSECTION WALL

TYP. CONNECTION DETAIL OF MASONRY WALL

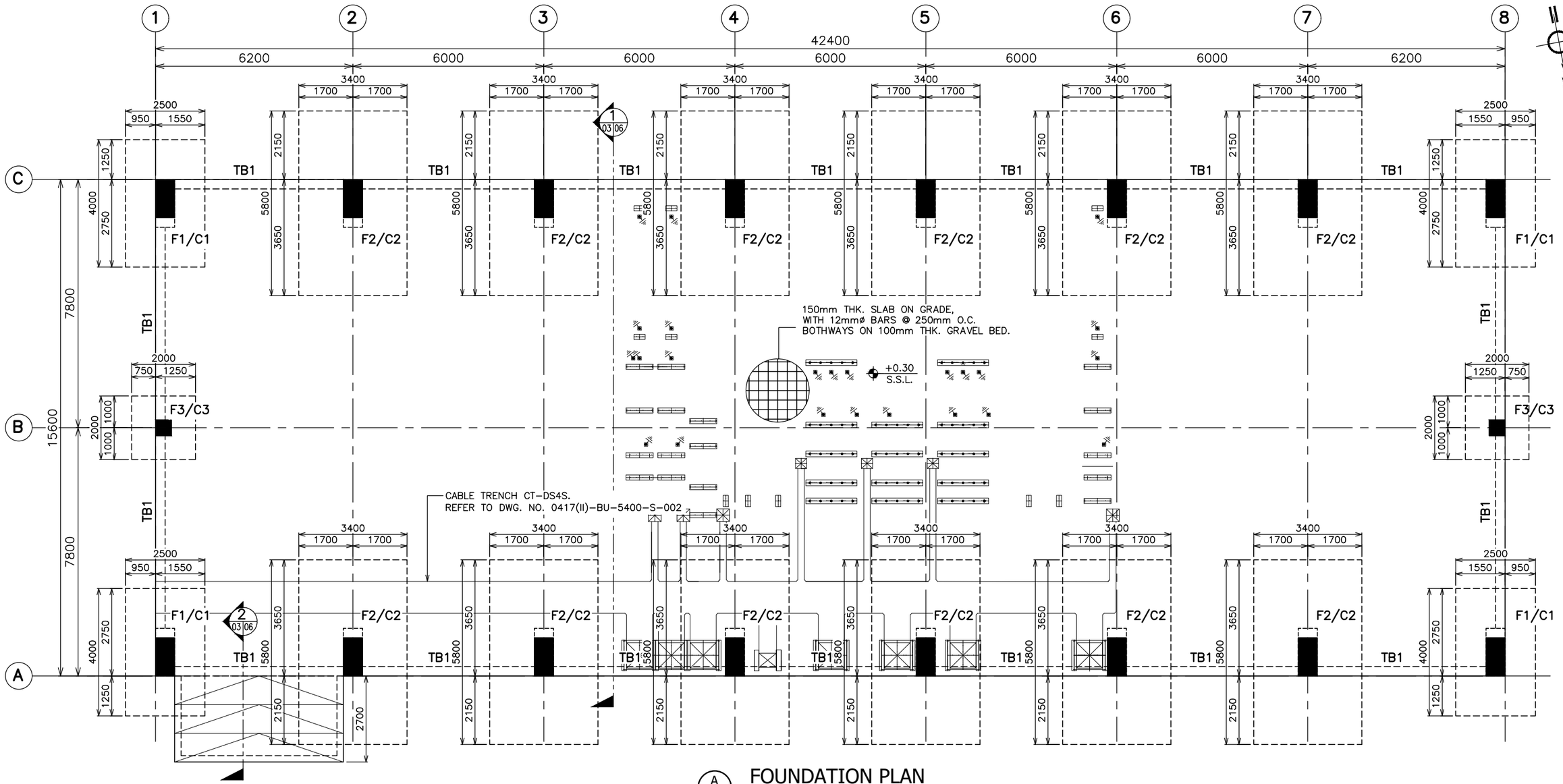
SCALE : NTS



SECTION

PIPE SLEEVE THRU WALL

SCALE: NTS



A 03 03 FOUNDATION PLAN

SCALE 1:130

NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

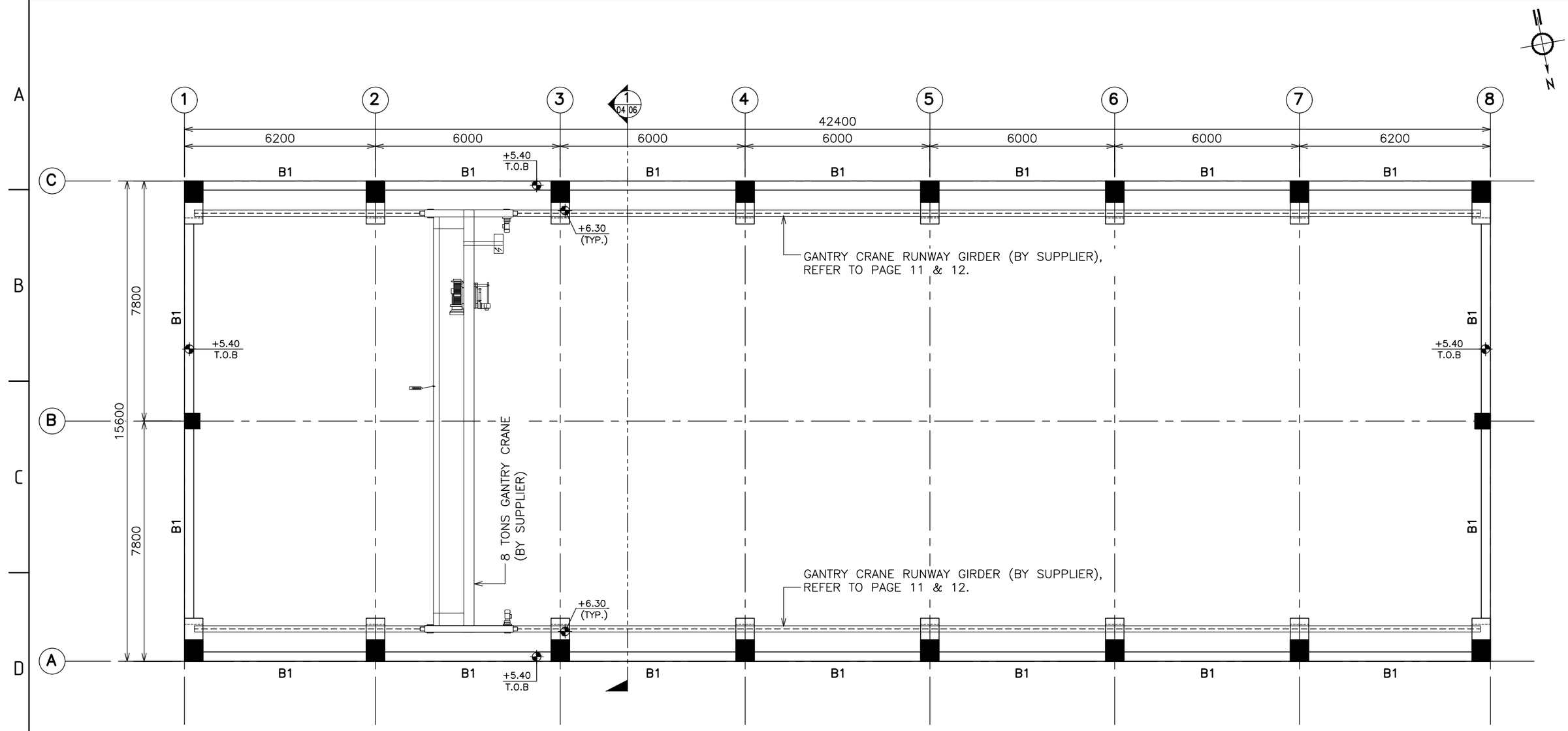
1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. U.O.S., ALL CHB WALL SHALL REST ON TIE BEAMS.
3. THE DESIGN NET ALLOWABLE SOIL BEARING CAPACITY USED IS 434 kPa AT MINIMUM FOUNDING LEVEL OF 3.00m BELOW NATURAL GRADE BASED ON BOREHOLE BH-03 OF GEOTECHNICAL REPORT.
4. SITE ADAPTATION: THE GENERAL CONTRACTOR SHALL INFORM THE STRUCTURAL ENGINEER OF ANY SITE CONDITION THAT MAY REQUIRE CHANGES IN THE STRUCTURAL DESIGN.
5. FOR FOUNDATION DETAILS & SECTIONS REFER TO SHEET NO. 06
6. FOR TIE BEAMS REINFORCEMENT, REFER TO BEAM SCHEDULE AT SHEET NO. 08
7. REFER TO SITE DEVELOPMENT PLAN AND CIVIL WORKS DRAWINGS FOR THE DETAILS OF CABLE TRENCH.
8. FOR FOUNDATION OF 138kV GIS BUSHING SUPPORT, REFER TO DWG. NO. 0417(II)-BU-5440-S-004
9. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.

LEGEND:

TB1	TIE BEAM MARK
F1/C1	FOOTING/ COLUMN MARK
S.S.L.	STRUCTURAL SLAB LEVEL
F.G.L.	FINISH GRADE LINE
N.G.L.	NATURAL GRADE LEVEL
R.L.	ROAD LEVEL

A 01 02 ← REFER TO DWG. NO.

↑ DWG. NO.



B 04 04 PLAN @ LEVEL +6.30

SCALE 1:130

NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. FOR BEAM REINFORCEMENT, REFER TO BEAM SCHEDULE AT SHEET NO. 08
3. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.
4. **SUPPLY, FABRICATION AND INSTALLATION OF GANTRY CRANE RUNWAY GIRDER INCLUDING THE SUPPORT CONNECTION ARE SCOPE OF THE SUPPLIER**

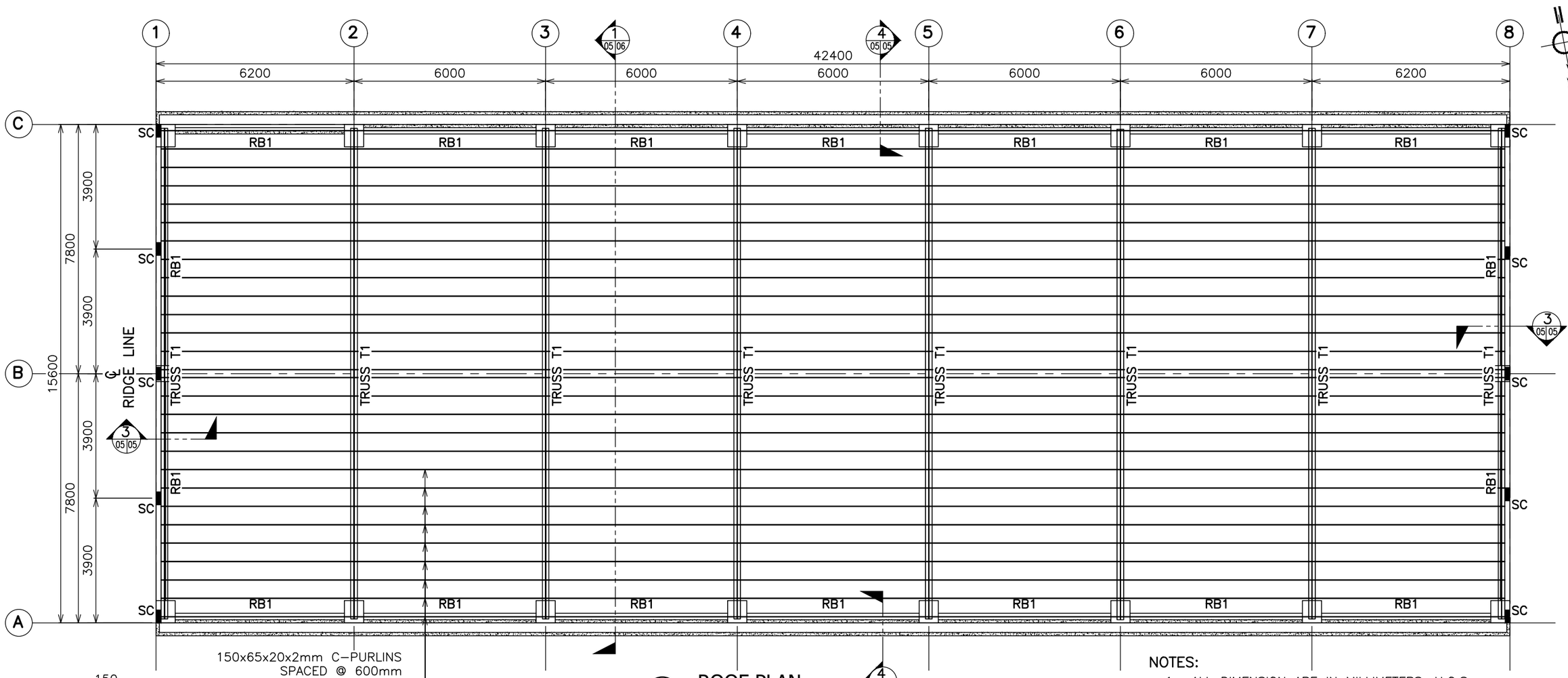
LEGEND:

B1 BEAM MARK

T.O.B. TOP OF BEAM LEVEL

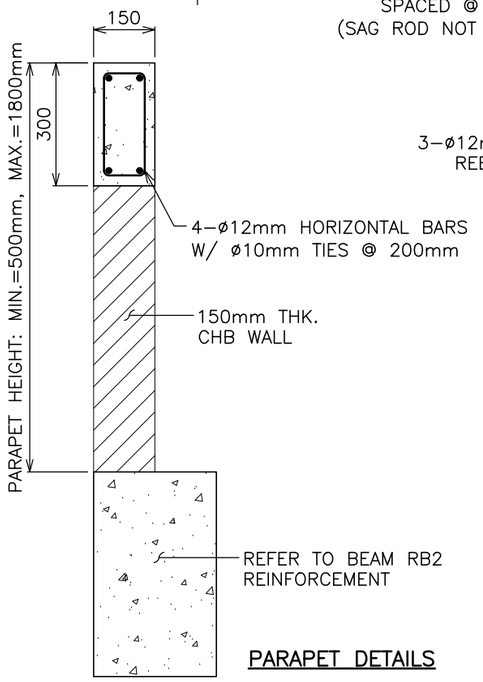
A 01 02 ← REFER TO DWG. NO.

↑ DWG. NO.



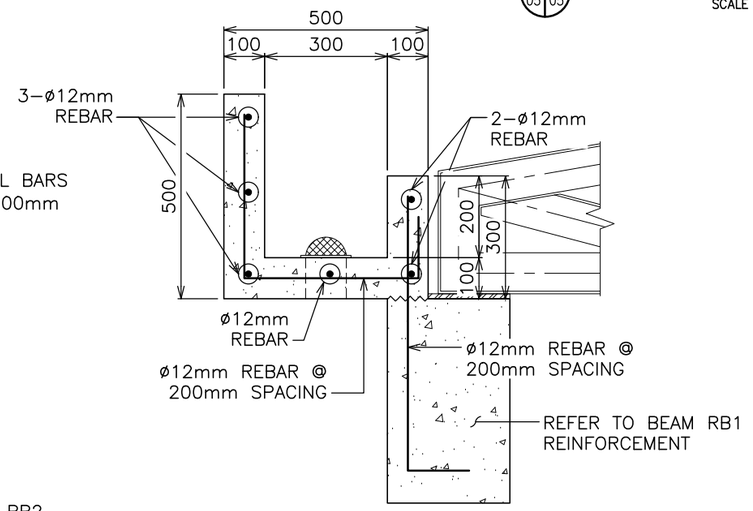
150x65x20x2mm C–PURLINS
SPACED @ 600mm
(SAG ROD NOT SHOWN)

C 05 05 ROOF PLAN
SCALE 1:130



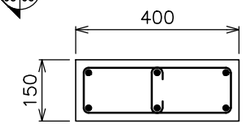
PARAPET DETAILS

D 05 05 SECTION 3
SCALE 1:15



RC. GUTTER DETAILS

E 05 05 SECTION 4
SCALE 1:15



COLUMN SC

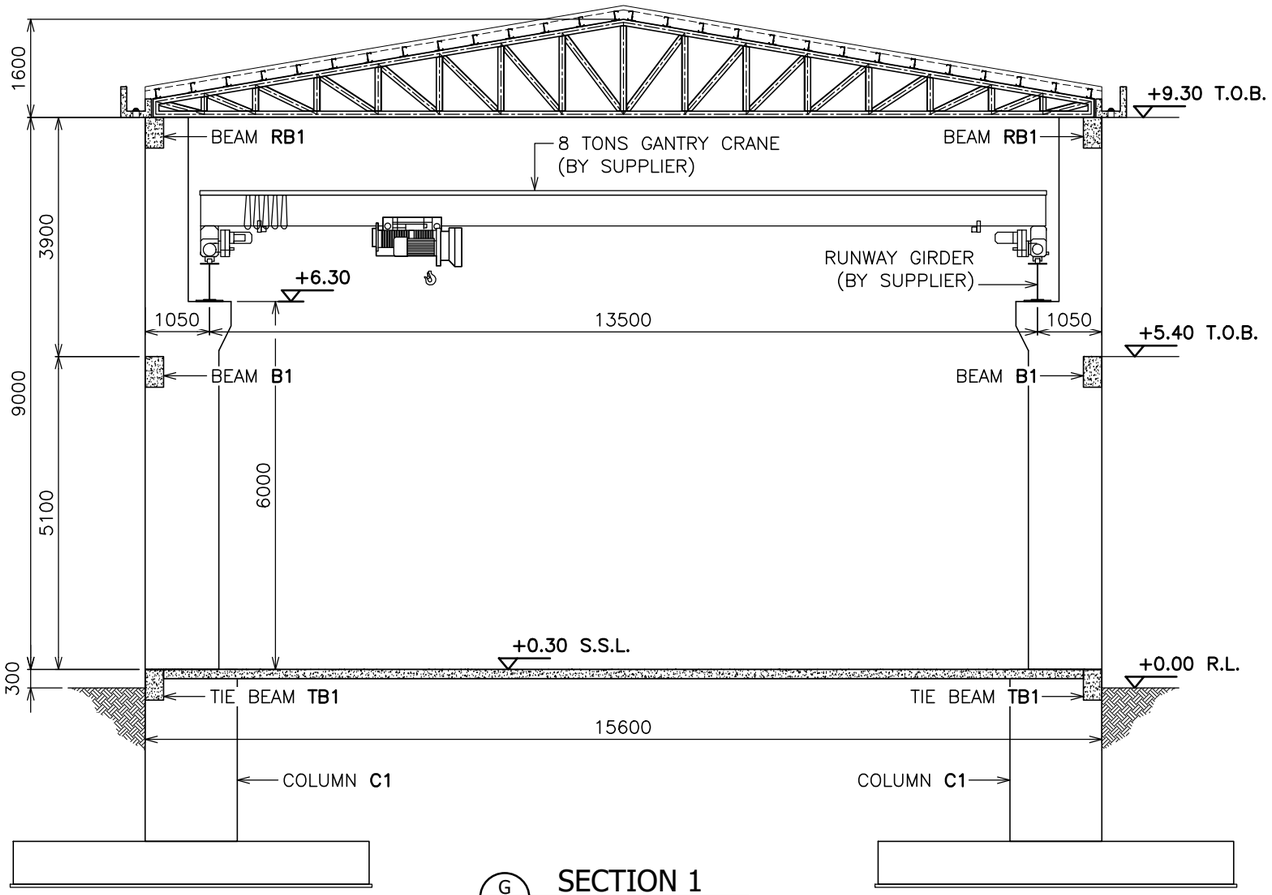
REINFORCEMENT:

6–ø12mm VERTICAL BARS
W/ ø10mm TIES @ 150mm
O.C. (2/SET)

F 05 05 PARAPET COLUMN- SC
SCALE 1:15

NOTES:

1. ALL DIMENSION ARE IN MILLIMETERS, U.O.S.
2. REFER TO GENERAL NOTES FOR THE REINFORCEMENT OF 150mm THK. CHB WALL.
3. U.O.S., ALL STRUCTURAL STEEL MEMBERS SHALL CONFORM TO ASTM SPECIFICATIONS FOR A–36 STEEL.
4. PROVIDE SAG RODS AT EVERY MIDDLE THIRD.
5. VERIFY ALL LENGTH OF TRUSSES AS PER SITE ACTUAL DIMENSION.
6. REFER TO PAGE 9 & 10 FOR ROOF & TRUSS DETAILS
7. ROOF SLOPE IS EQUAL TO 10 DEGREES.
8. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.

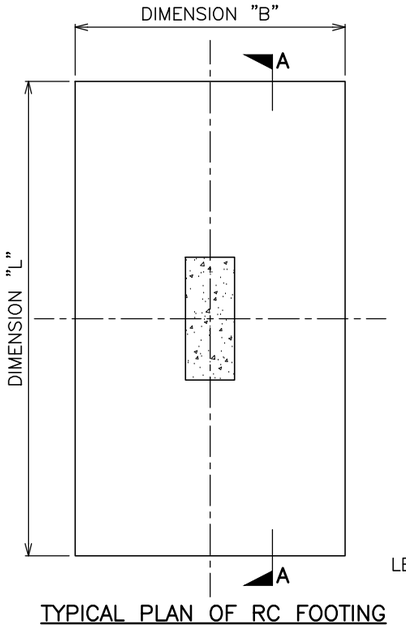


G 06|03 SECTION 1

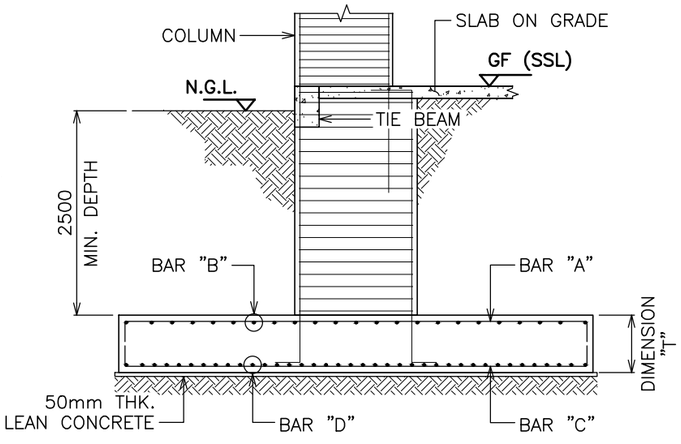
SCALE 1:100

FOOTING SCHEDULE

FOOTING MARK	FOOTING DIMENSION (mm)			TOP REINFORCEMENT		BOTTOM REINFORCEMENT	
	DIM. "B"	DIM. "L"	DIM. "T"	BAR "A"	BAR "B"	BAR "C"	BAR "D"
F1	2500	4000	600	ø20 @ 150mm	ø20 @ 150mm	ø20 @ 150mm	ø20 @ 150mm
F2	3400	5800	800	ø20 @ 150mm	ø20 @ 150mm	ø20 @ 150mm	ø20 @ 150mm
F3	2000	2000	400	ø16 @ 150mm	ø16 @ 150mm	ø16 @ 150mm	ø16 @ 150mm



TYPICAL PLAN OF RC FOOTING



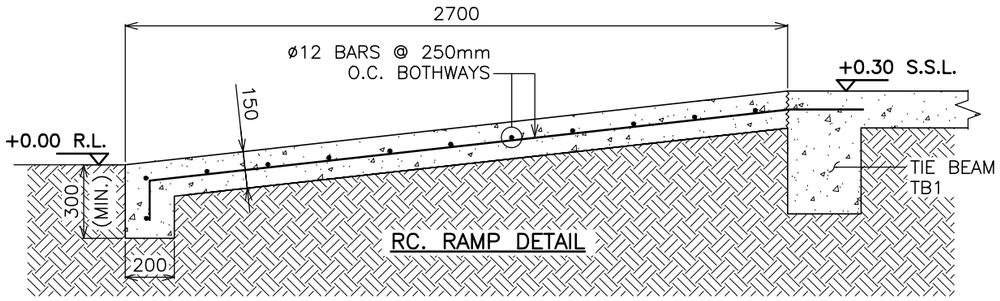
SECTION A–A

H 06|03 TYPICAL FOUNDATION DETAILS

SCALE NTS

NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. U.O.S., ALL CHB WALL SHALL REST ON TIE BEAMS.
3. THE DESIGN NET ALLOWABLE SOIL BEARING CAPACITY USED IS 434 kPa AT MINIMUM FOUNDING LEVEL OF 3.00m BELOW NATURAL GRADE BASED ON BOREHOLE BH–03 OF GEOTECHNICAL REPORT.
4. SITE ADAPTATION: THE GENERAL CONTRACTOR SHALL INFORM THE STRUCTURAL ENGINEER OF ANY SITE CONDITION THAT MAY REQUIRE CHANGES IN THE STRUCTURAL DESIGN.
5. FOR TIE BEAMS REINFORCEMENT, REFER TO BEAM SCHEDULE AT SHEET NO. 08
6. REFER TO SITE DEVELOPMENT PLAN AND CIVIL WORKS DRAWINGS FOR THE DETAILS OF CABLE TRENCH.
7. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.



I 06|03 SECTION 2

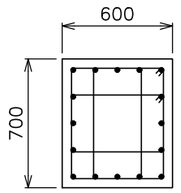
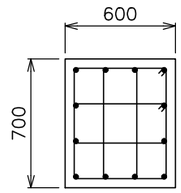
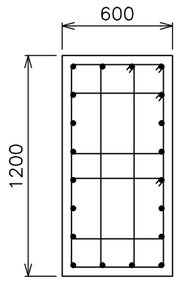
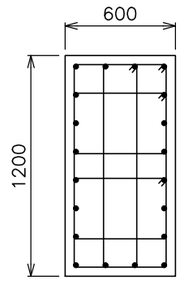
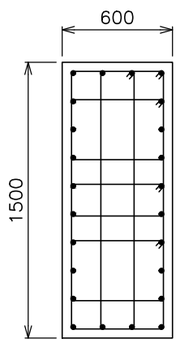
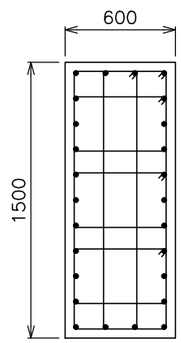
SCALE 1:25

LEGEND:

TB1	TIE BEAM MARK
F1/C1	FOOTING/ COLUMN MARK
S.S.L.	STRUCTURAL SLAB LEVEL
F.G.L.	FINISH GRADE LINE
N.G.L.	NATURAL GRADE LEVEL
R.L.	ROAD LEVEL

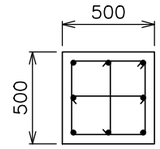
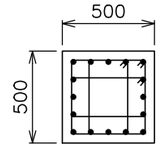
A 01|02 ← REFER TO DWG. NO.

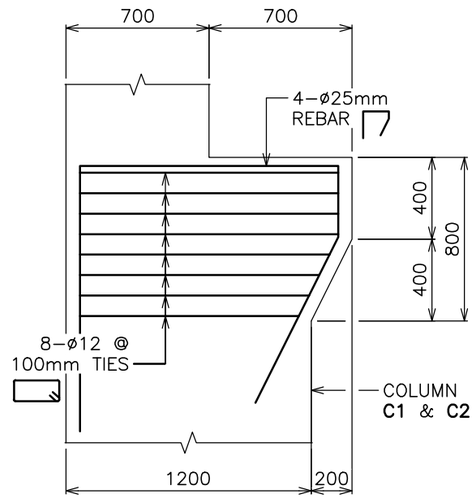
└ DWG. NO.

MARK / LEVEL	C1	C2
ROOF ↕ LEVEL +6.30		
	SIZE: 600 x 700	SIZE: 600 x 700
	VERT. BARS: 16–ø20	VERT. BARS: 12–ø32
	TIES: ø10 @ 200 O.C. (3/SETS)	TIES: ø12 @ 200 O.C. (3/SETS)
LEVEL +6.30 ↕ GROUND FLR.		
	SIZE: 600 x 1200	SIZE: 600 x 1200
	VERT. BARS: 20–ø25	VERT. BARS: 20–ø32
	TIES: ø10 @ 200 O.C. (4/SETS)	TIES: ø12 @ 200 O.C. (4/SETS)
GROUND FLR. ↕ FOUNDATION		
	SIZE: 600 x 1500	SIZE: 600 x 1500
	VERT. BARS: 24–ø25	VERT. BARS: 26–ø32
	TIES: ø10 @ 200 O.C. (5/SETS)	TIES: ø12 @ 200 O.C. (5/SETS)

COLUMN SCHEDULE

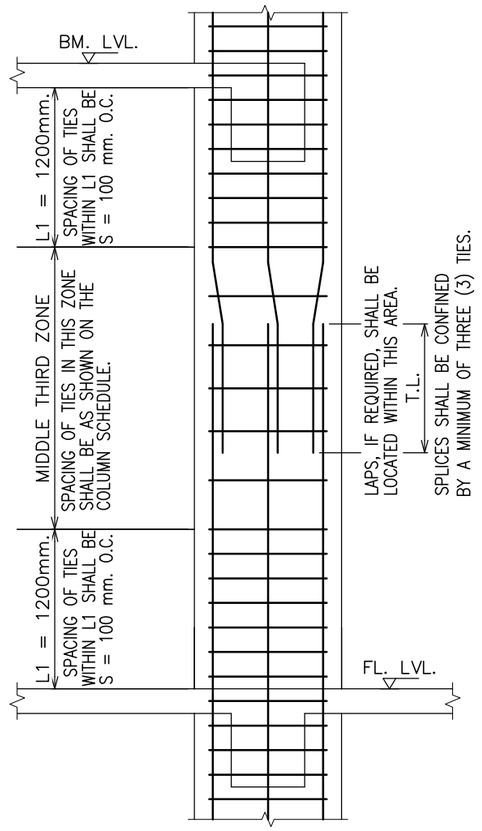
(REFER TO TYPICAL COLUMN DETAIL FOR TIE SPACING)

MARK / LEVEL	C3
ROOF ↕ GROUND FLR.	
	SIZE: 500 x 500
	VERT. BARS: 8–ø25
	TIES: ø10 @ 200 O.C. (3/SETS)
GROUND FLR. ↕ FOUNDATION	
	SIZE: 500 x 500
	VERT. BARS: 16–ø25
	TIES: ø10 @ 200 O.C. (3/SETS)

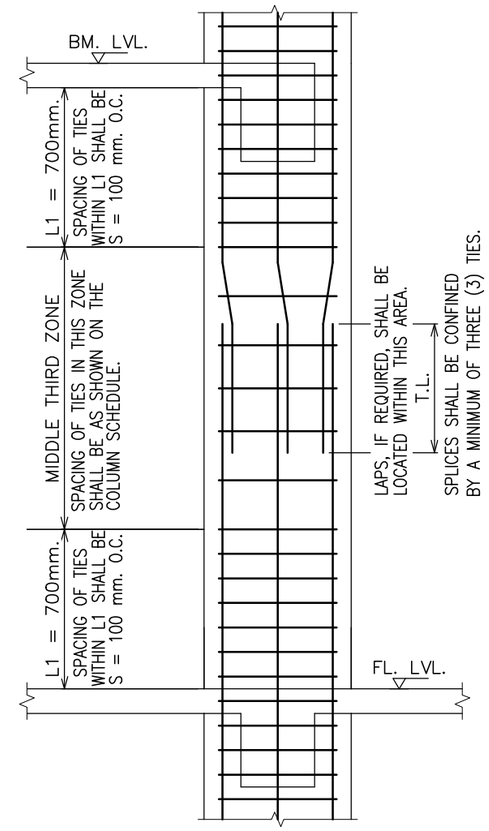


CORBEL DETAILS

SCALE NTS



COLUMN C1 & C2



COLUMN C3

LEGEND:

T.L. = FULL TENSION LAP SPLICE
PROVIDE TIES OVER LAP LENGTH
AT S=100mm.

TYPICAL COLUMN DETAILS

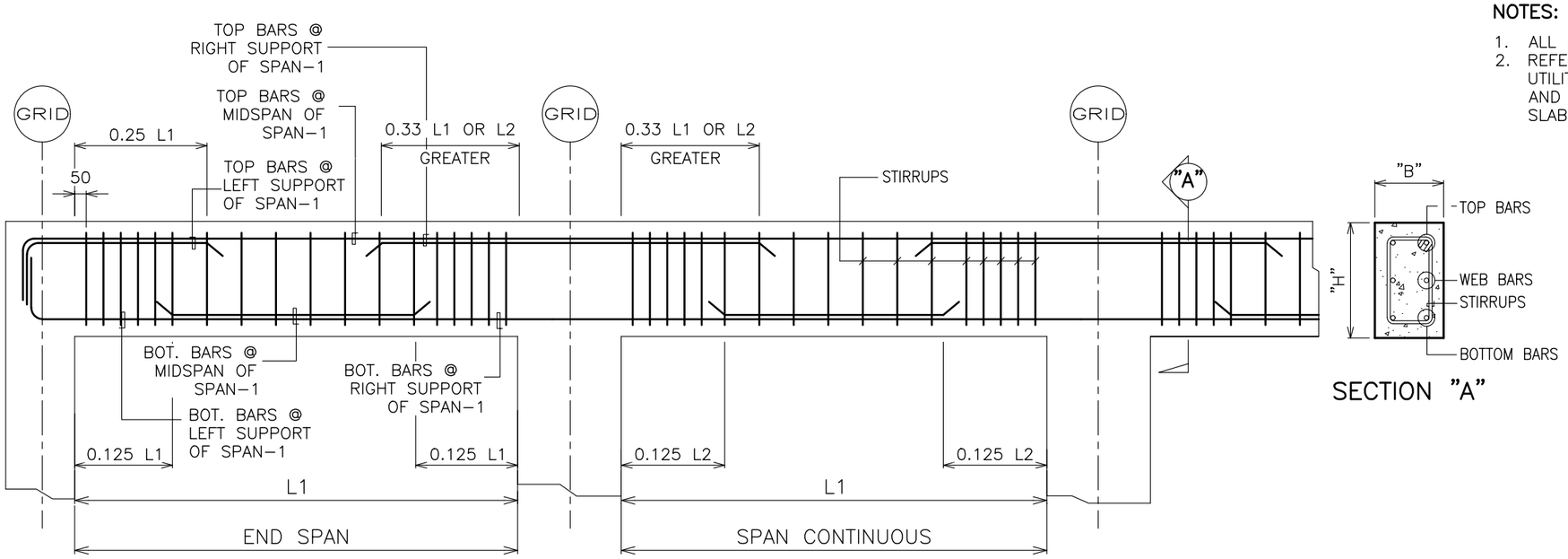
SCALE NTS

NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.

NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.



K 08|03 **TYPICAL BEAM DETAILS**

SCALE NTS

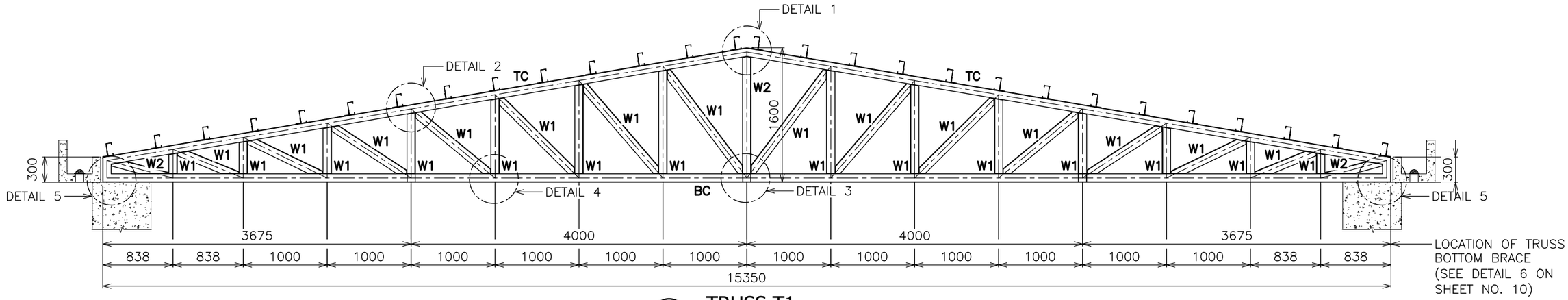
SECTION "A"

SCHEDULE OF STIRRUPS

TYPE A – 1 SET 10mmø, 1 @ 50, 20 @ 100mm O.C REST @ 200mm O.C

SCHEDULE OF BEAMS

BEAM MARK	SECTION		REINFORCEMENT						WEB BARS	STIRRUPS	REMARKS
			LEFT SUPPORT		MIDSPAN		RIGHT SUPPORT				
	B (mm)	H (mm)	TOP BAR	BOTTOM BAR	TOP BAR	BOTTOM BAR	TOP BAR	BOTTOM BAR			
TB1	300	500	3–25mmø	3–16mmø	3–25mmø	3–25mmø	3–25mmø	3–16mmø		TYPE–A	
B1	300	500	3–25mmø	3–16mmø	3–25mmø	3–25mmø	3–25mmø	3–16mmø		TYPE–A	
RB1	300	500	3–20mmø	3–16mmø	3–20mmø	3–20mmø	3–20mmø	3–16mmø		TYPE–A	

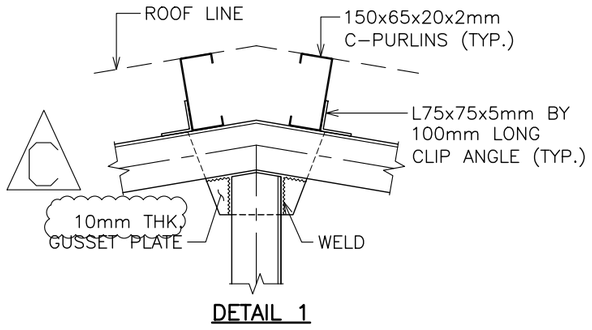


L 09|05 **TRUSS T1**

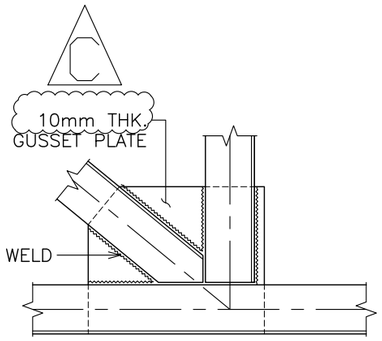
SCALE 1:50

TRUSS MEMBERS:

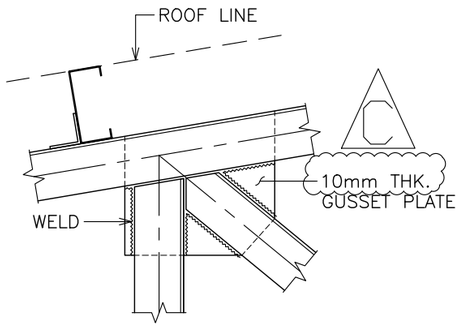
TC = 2L100x100x6mm ANGLE BAR (TOP CHORD)
BC = 2L100x100x6mm ANGLE BAR (BOTTOM CHORD)
W1 = L100x100x6mm ANGLE BAR (WEB MEMBER)
W2 = 2L100x100x6mm ANGLE BAR (WEB MEMBER)



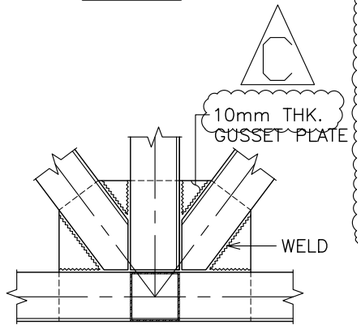
DETAIL 1



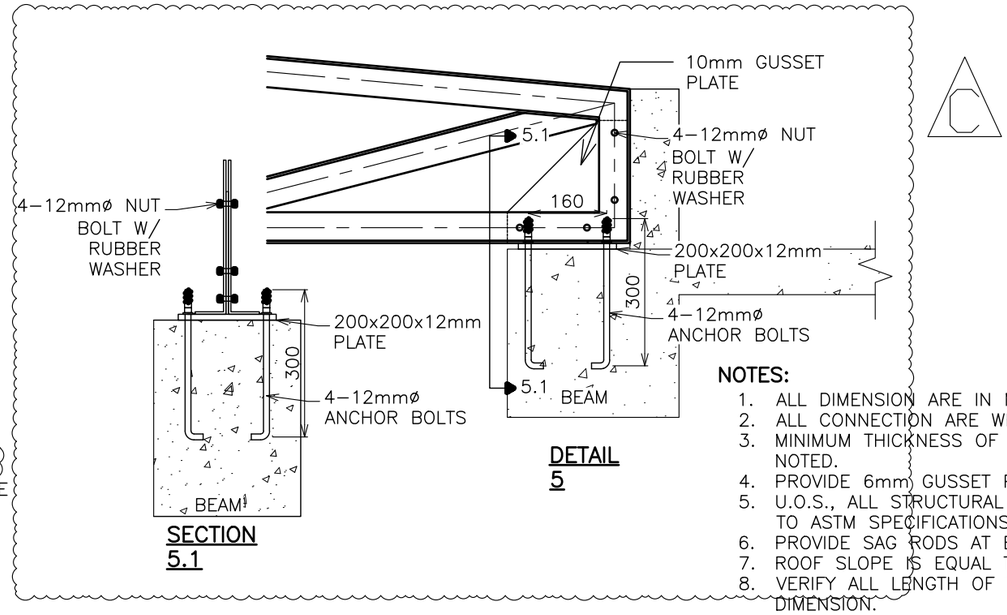
DETAIL 4



DETAIL 2



DETAIL 3



SECTION 5.1

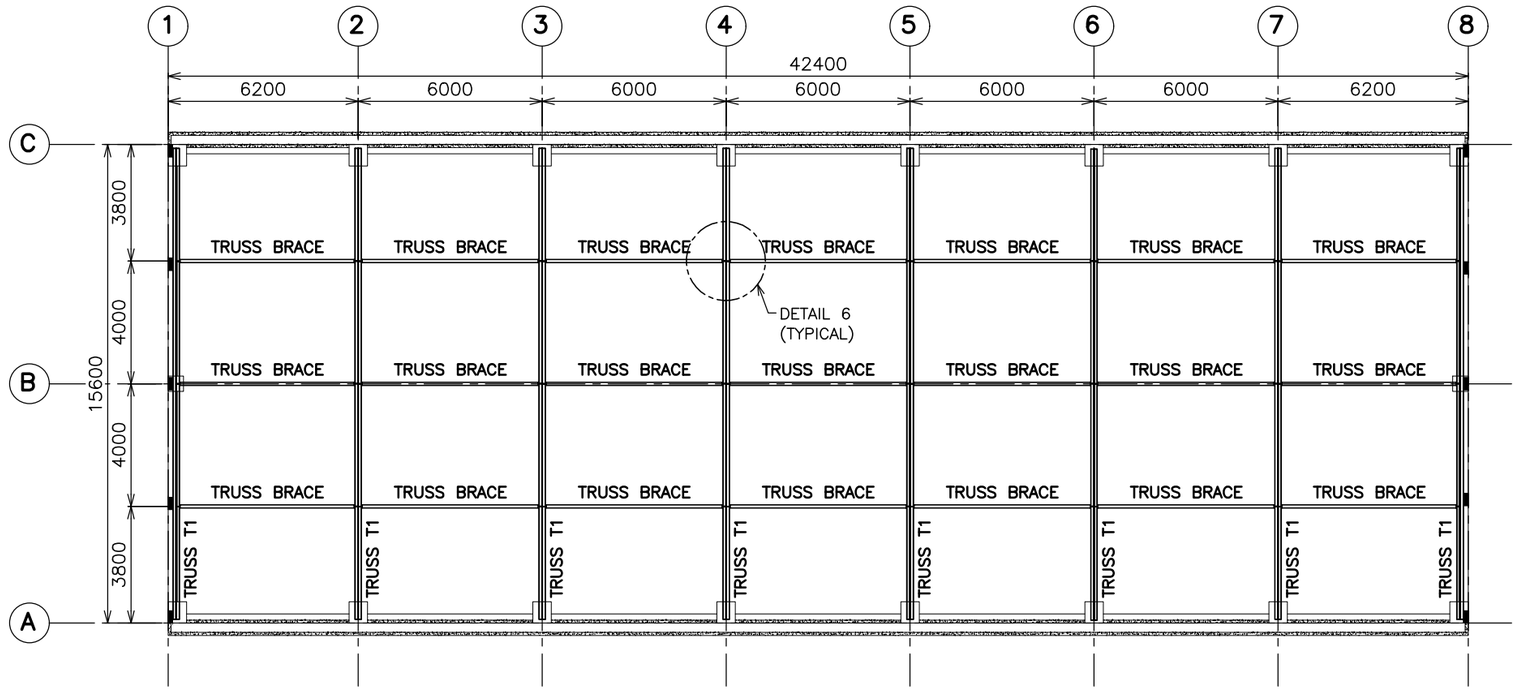
DETAIL 5

NOTES:

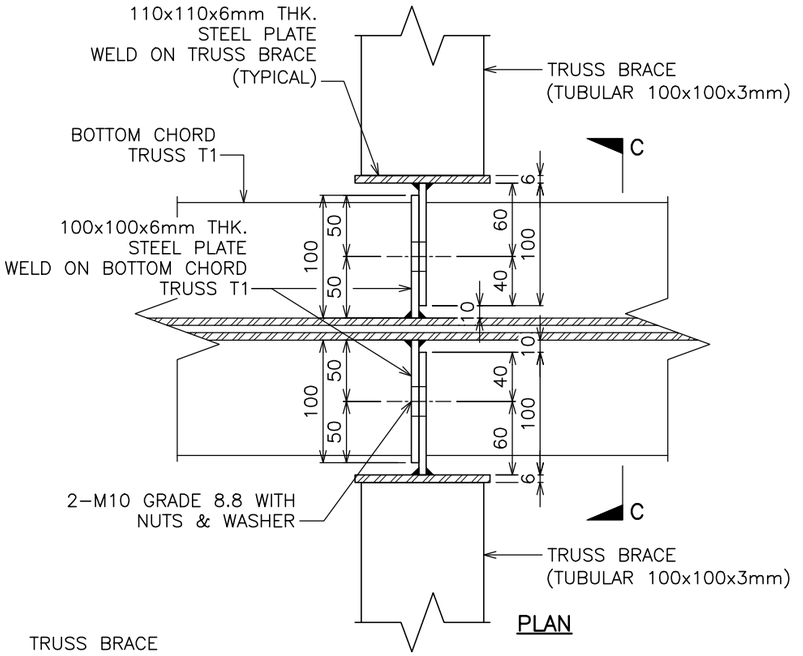
1. ALL DIMENSION ARE IN MILLIMETERS, U.O.S.
2. ALL CONNECTION ARE WELDED UNLESS OTHERWISE NOTED.
3. MINIMUM THICKNESS OF WELD = 6mm UNLESS OTHEWISE NOTED.
4. PROVIDE 6mm GUSSET PLATE IN ALL TRUSS CONNECTION.
5. U.O.S., ALL STRUCTURAL STEEL MEMBERS SHALL CONFORM TO ASTM SPECIFICATIONS FOR A-36 STEEL.
6. PROVIDE SAG RODS AT EVERY MIDDLE THIRD.
7. ROOF SLOPE IS EQUAL TO 10 DEGREES.
8. VERIFY ALL LENGTH OF TRUSSES AS PER SITE ACTUAL DIMENSION.

M 09|09 **TYPICAL DETAILS OF TRUSS**

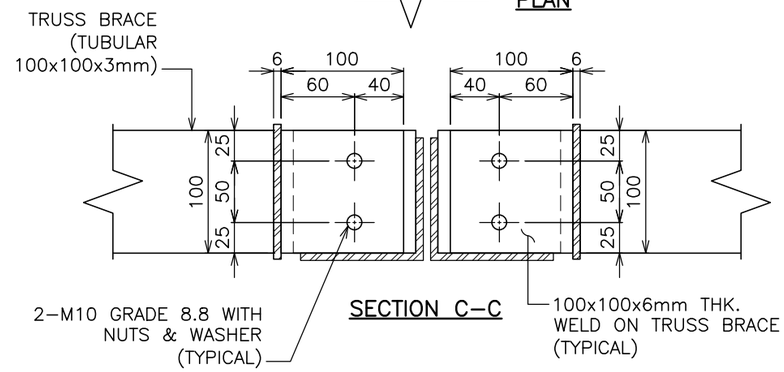
SCALE NTS



BOTTOM TRUSS PLAN
SCALE 1:200

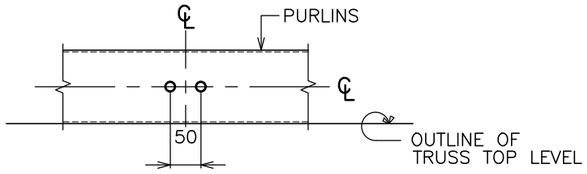


PLAN

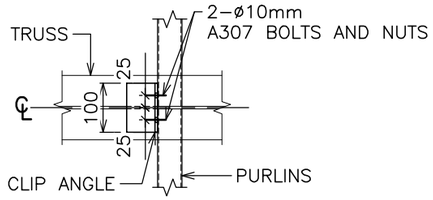


SECTION C–C

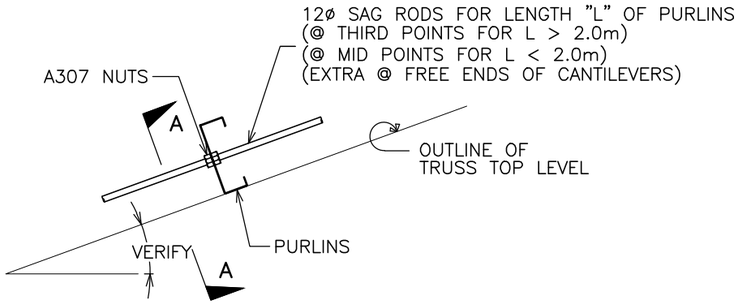
TYPICAL DETAIL OF TRUSS
BOTTOM BRACE CONNECTION



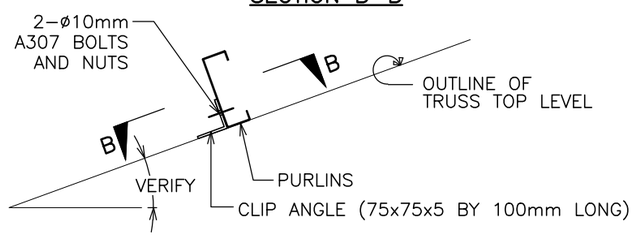
SECTION A–A



SECTION B–B



TYPICAL DETAIL OF SAG RODS
SCALE NTS

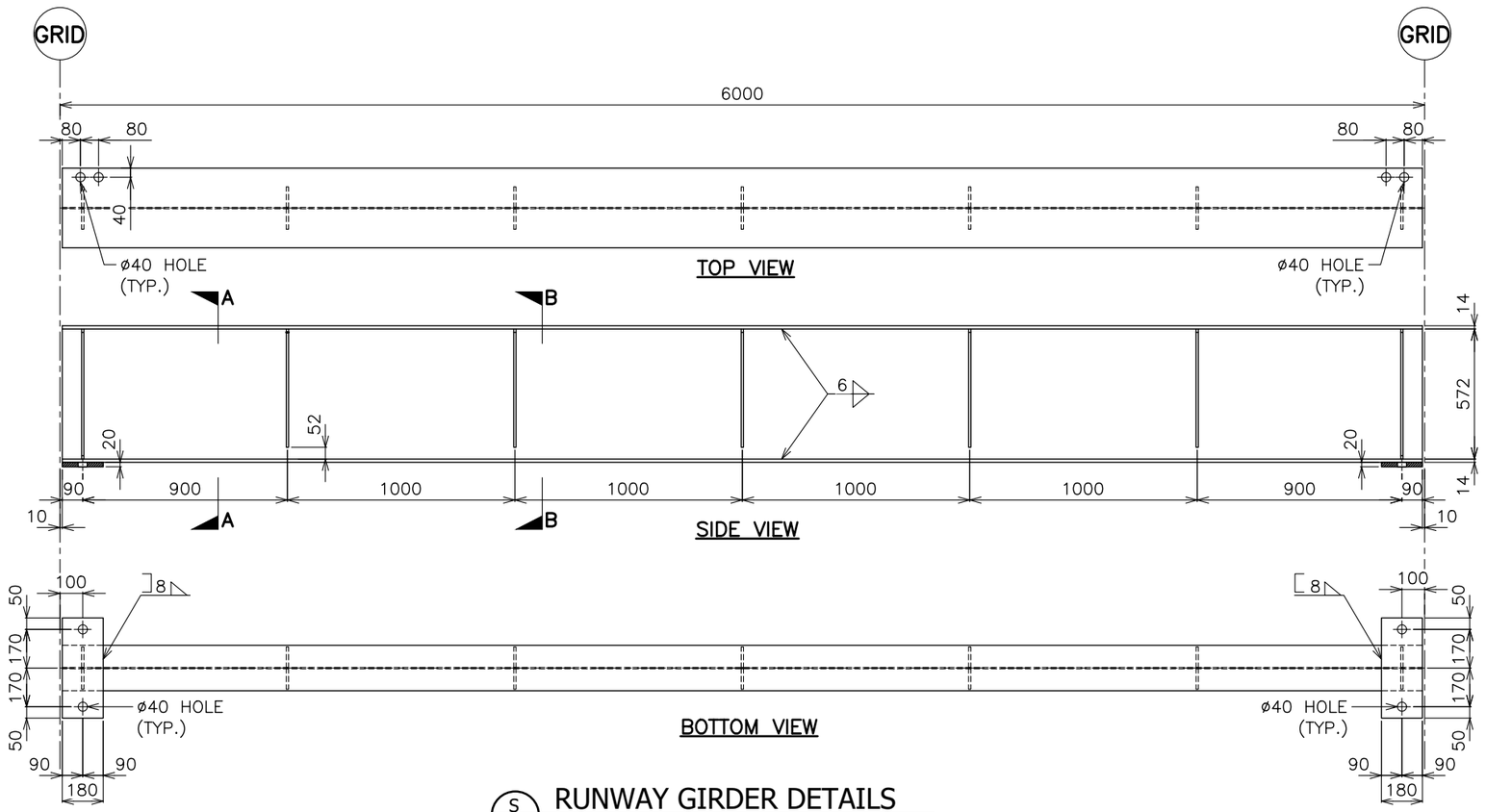


PURLIN TO TRUSS CONNECTION DETAILS
SCALE NTS

DETAIL 6
SCALE NTS

NOTES:

1. ALL DIMENSION ARE IN MILLIMETERS, U.O.S.
2. ALL CONNECTION ARE WELDED UNLESS OTHERWISE NOTED.
3. MINIMUM THICKNESS OF WELD = 3mm UNLESS OTHEWISE NOTED.
4. U.O.S., ALL STRUCTURAL STEEL MEMBERS SHALL CONFORM TO ASTM SPECIFICATIONS FOR A–36 STEEL.
5. PROVIDE SAG RODS AT EVERY MIDDLE THIRD.
6. ROOF SLOPE IS EQUAL TO 10 DEGREES.
7. VERIFY ALL LENGTH OF TRUSSES AS PER SITE ACTUAL DIMENSION.



RUNWAY GIRDER DETAILS

S | 10 | 04

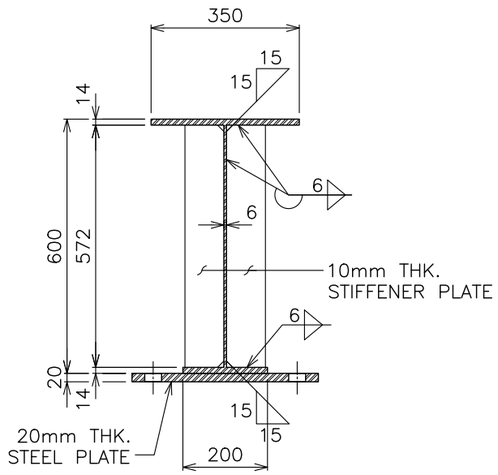
SCALE 1:25

NOTES:

1. ALL DIMENSION ARE IN MILLIMETERS, LEVELS AND COORDINATES ARE IN METERS, U.O.S.
2. THE CONTRACTOR HAS TO CHECK ALL DIMENSIONS, LEVELS, COORDINATES AND ACTUAL SITE CONDITION.
3. ANY DISCREPANCIES SHALL BE REPORTED AND CLARIFIED PRIOR TO COMMENCING OF WORK.
4. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT DRAWINGS AND SPECIFICATIONS.
5. MATERIALS:

STEEL PLATES/SECTIONS:	Q235
STEEL BOLTS:	GRADE 8.8
WELDING:	E70XX

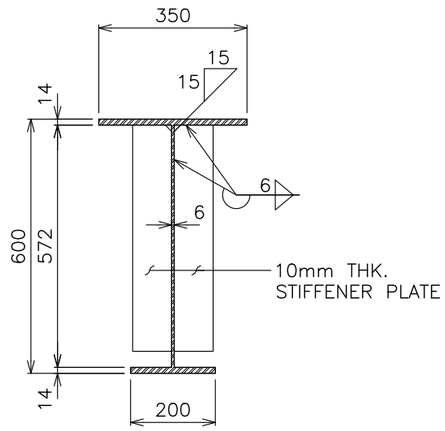
6. THE STRUCTURE PAINTING SHOULD FOLLOW THE PROJECT GENERAL REQUIREMENT. BEFORE PAINTING, THE STEEL SURFACE OF STEEL COMPONENTS SHALL BE SANDBLASTED (BLASTING) FOR DERUSTING AND MEET THE REQUIREMENT OF CLASS–SA2.5. THE STEEL PAINTING: EPOXY ZINC RICH PRIMER 75UM EPOXY MICACEOUS IRON INTERMEDIATE PAINT 75UM ALIPHATIC POLYURETHANE FINISH 70UM, ALMOST WINTER TOP COATING.
7. **SUPPLY, FABRICATION AND INSTALLATION OF GANTRY CRANE RUNWAY GIRDER INCLUDING THE SUPPORT CONNECTION ARE SCOPE OF THE SUPPLIER.**



T | 10 | 10

SECTION A-A

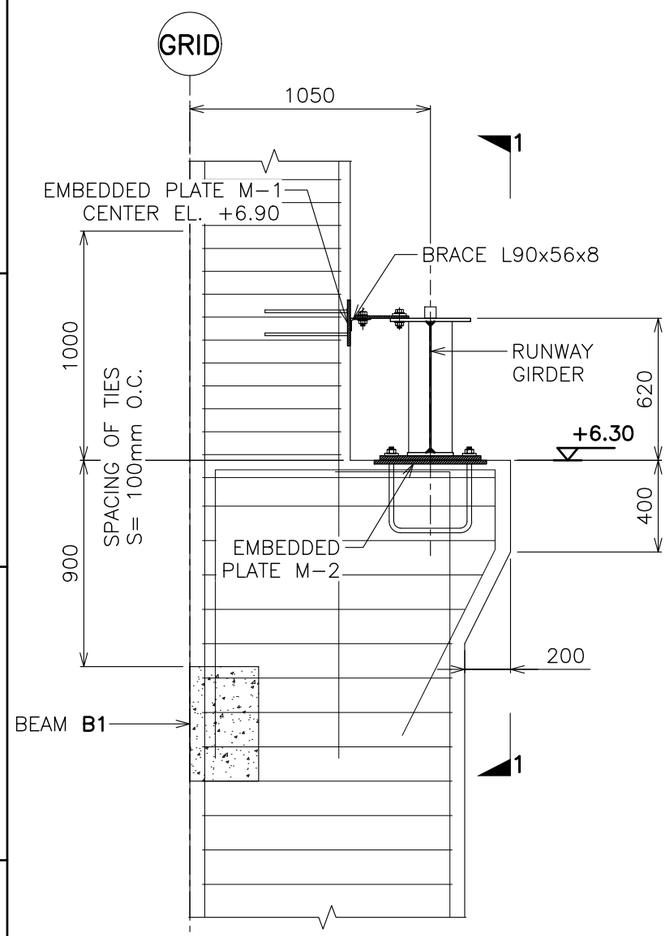
SCALE 1:15



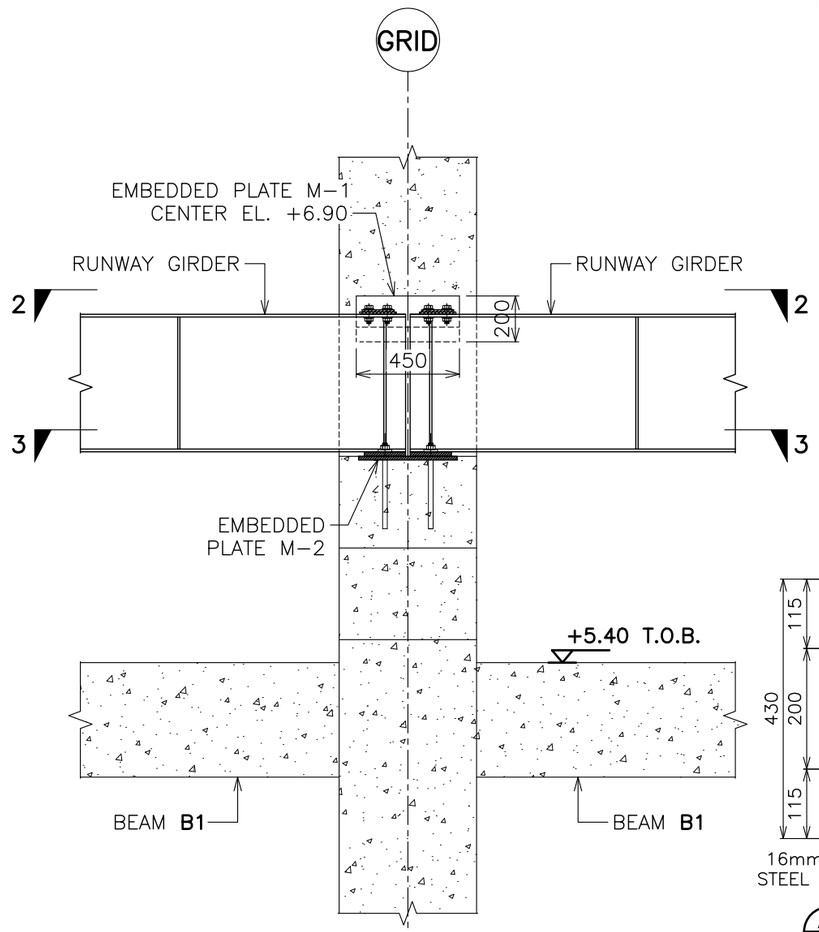
U | 10 | 10

SECTION B-B

SCALE 1:15



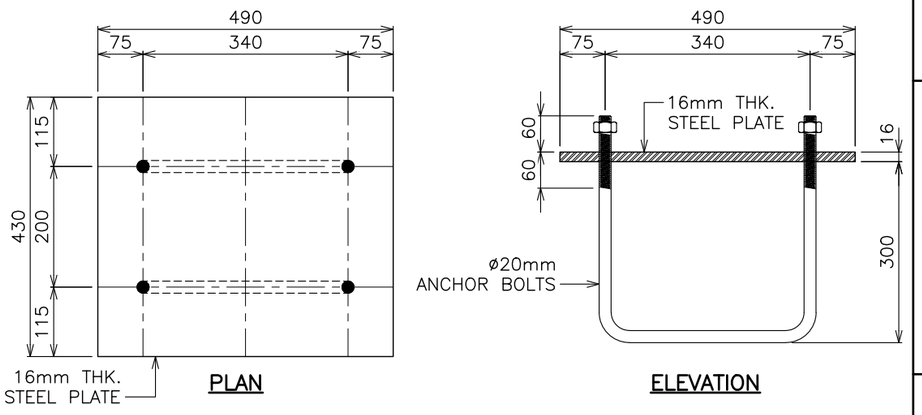
V 12|06 RUNWAY GIRDER DETAILS
SCALE 1:25



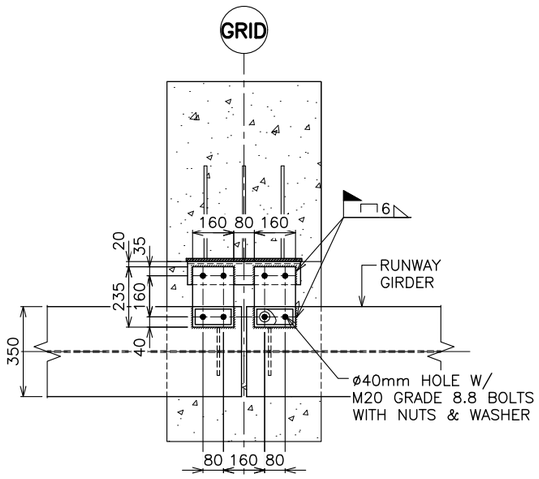
W 12|12 SECTION 1-1
SCALE 1:25

NOTES:

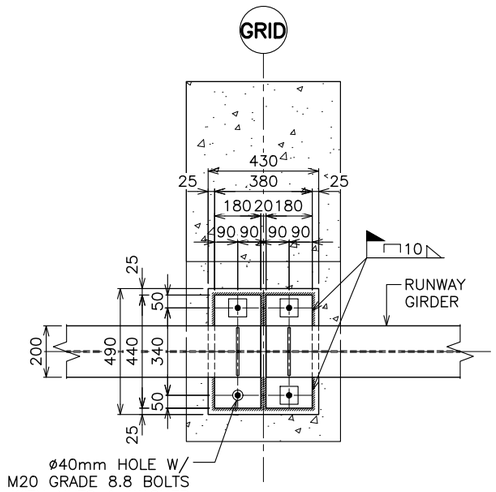
1. ALL DIMENSION ARE IN MILLIMETERS, LEVELS AND COORDINATES ARE IN METERS, U.O.S.
2. THE CONTRACTOR HAS TO CHECK ALL DIMENSIONS, LEVELS, COORDINATES AND ACTUAL SITE CONDITION.
3. ANY DISCREPANCIES SHALL BE REPORTED AND CLARIFIED PRIOR TO COMMENCING OF WORK.
4. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT DRAWINGS AND SPECIFICATIONS.
5. MATERIALS:
   - STEEL PLATES/SECTIONS: Q235
   - STEEL BOLTS: GRADE 8.8
   - WELDING: E70XX
6. THE STRUCTURE PAINTING SHOULD FOLLOW THE PROJECT GENERAL REQUIREMENT. BEFORE PAINTING, THE STEEL SURFACE OF STEEL COMPONENTS SHALL BE SANDBLASTED (BLASTING) FOR DERUSTING AND MEET THE REQUIREMENT OF CLASS–SA2.5. THE STEEL PAINTING: EPOXY ZINC RICH PRIMER 75UM EPOXY MICACEOUS IRON INTERMEDIATE PAINT 75UM ALIPHATIC POLYURETHANE FINISH 70UM, ALMOST WINTER TOP COATING.
7. **SUPPLY, FABRICATION AND INSTALLATION OF GANTRY CRANE RUNWAY GIRDER INCLUDING THE SUPPORT CONNECTION ARE SCOPE OF THE SUPPLIER.**



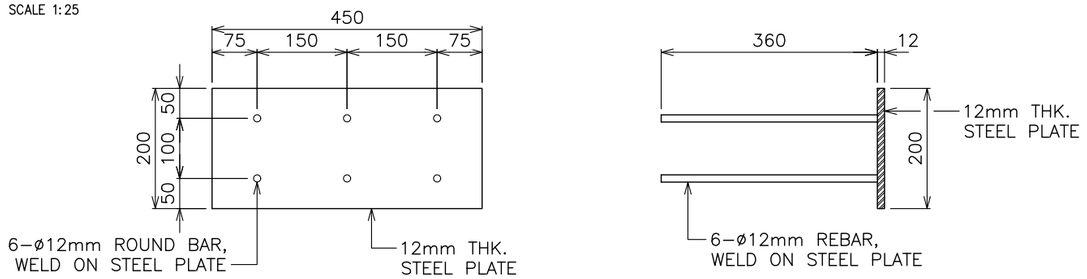
A1 12|12 EMBEDDED PLATE M-2
SCALE 1:10



X 12|12 SECTION 2-2
SCALE 1:20



Y 12|12 SECTION 3-3
SCALE 1:20



Z 12|12 EMBEDDED PLATE M-1
SCALE 1:10